Research Statement

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My current research is mainly in quantum information theory, at times with a tilt towards quantum foundations. Specifically, I care about how quantum theory fundamentally constrains or enables information processing in a precise sense, and what the resulting limits reveal about the operational structure of quantum theory itself. The purpose of research, for me, is to understand how nature works and hopefully to draw lessons instructive for navigating everyday life. Technical discovery, conceptual insight, as well as thoughtful interpretation and presentation are all integral parts of my notion of ideal theoretical research.

1 Past works

My past works can be classified into the following themes.

1.1 Quantum Shannon theory and exotic resources

One of the core conceptual ideas of quantum Shannon theory is that communication is conducted on physical substrates, and as a result, the ultimate limit, and even the very objectives, of communication can vary significantly depending on the physical nature of the communication medium and on the availability of resource assistance allowed by the underlying physical theory. For instance, communicating over a noisy quantum channel instead of a classical one opens up the possibility of enhancing the ultimate limit of communication with entanglement assistance, and it unlocks new objectives of communication beyond the transmission of classical messages: one can now consider transmitting quantum messages, or a hybrid of the two. One theme of my past works has been to carry this idea even further, in particular beyond the standard quantum formalism.

In my work with Lloyd and Wilde [1], we developed a theory of communication in which the communication medium is a noisy quantum closed timelike curve (CTC), namely a noisy mechanism that sends substances backward in time (as opposed to forward in time, as conventional channels do). While CTCs do not exist within the standard quantum formalism, they are in principle permitted by general relativity and may arise in the vicinity of black holes according to final-state projection models [2, 3]. With noisy CTCs as communication media, the natural objective is to transmit classical or quantum messages backward in time, from a sender in the future to a receiver in the past. We call this task *retrocausal communication*, and its maximum achievable efficiency the retrocausal capacity of the underlying noisy CTC. Interestingly, this communication setup is a faithful portrayal of the climactic scene of the science-fiction film *Interstellar*, where the protagonist, inside a black hole, finds himself capable of sending messages through a bookshelf (a noisy CTC) to his childhood daughter in the past. Our main technical result is an exact analytical characterization of the one-shot retrocausal capacity of an arbitrary noisy CTC, as well as a regularized formula for the asymptotic retrocausal capacity, with an expression reminiscent of Shannon's mutual information. This result can be viewed as a coding theorem for a universe admitting (even noisy) time travel, and it demonstrates that the design principles of optimal communication protocols would be fundamentally different in such a universe.

In my work with Regula and Wilde [4], we investigated the ultimate limit of communication in a scenario where the receiver has the ability to perform *postselection*, namely the ability to force any event that occurs with nonzero probability to occur with probability one. Without bending the standard quantum formalism, postselection can be experimentally simulated by discarding all trials in which the desired event did not occur. In a communication setup, this amounts to providing the receiver with an option of inconclusive decoding and evaluating the accuracy of decoding only on the conclusive trials. Such a scenario is of particular practical relevance in high-stakes situations where errors are far more costly than abstention. We exactly determined the one-shot and asymptotic postselected capacities of an arbitrary quantum channel in entanglement-assisted communication. This serves as the counterpart of the entanglement-assisted capacity theorem [5] in the postselected setting.

One way to gain novel insight into a given theory is to alter some part of it and examine how the alteration changes conclusions known to hold in the original theory. In the same spirit, I hope that my work contributes some nontraditional insight, through quantum Shannon theory (as previous works did through computational complexity theory [6, 7]), to the understanding of quantum theory itself, and more broadly, to foundational topics in physics such as nonlinearity, time symmetry, indefinite causal structure, and quantum gravity.

1.2 Quantum hypothesis testing

Quantum hypothesis testing is one of the most fundamental tasks in quantum information. In binary quantum hypothesis testing, the goal is to guess the true identity of an unknown quantum state (or of i.i.d. copies thereof, in the asymptotic regime) given two known hypotheses. There are two types of errors that one can make: identifying the second hypothesis when the first is true, and vice versa. The optimal asymptotic performance of binary quantum hypothesis testing has been believed to be characterized by the quantum Hoeffding bound [8, 9, 10] and the quantum Chernoff bound [11, 12], with the former governing the optimal tradeoff between the exponents of the two types of errors and the latter governing the optimal exponent of the Bayesian average error.

In my work with Regula [13], we argued that these bounds can nevertheless be sidestepped for many practical purposes, merely at the cost of a negligible operational overhead. Specifically, this is made possible by employing protocols that may be *inconclusive* (i.e., identifying neither hypothesis) with a certain probability. In practical applications that demand a definite conclusion, the protocol can simply be repeated until conclusive, and the error exponents are evaluated with respect to the average total number of copies consumed across all repetitions. Note that such an approach is fundamentally different from a postselected one, as inconclusive trials are not discarded but counted within the analysis. Using an adaptation of a sequential protocol from Ref. [14], we showed that both types of errors in binary quantum hypothesis testing can decay according to the Stein exponents simultaneously, even with the inconclusive probability being vanishingly small, thus breaking through the tradeoff imposed by the quantum Hoeffding bound. Likewise, the quantum Chernoff bound can also be exceeded, despite the additional overhead incurred by inconclusiveness, when the Bayesian priors are uniform. We further showed that such improvements over the conventional bounds are (i) robust, in the sense that they persist (albeit to a smaller degree) even if the inconclusive probability vanishes exponentially fast, and (ii) strongly optimal, in the sense that demanding larger improvements necessarily forces the inconclusive probability to approach one. We also exactly determined the optimal rates at which the inconclusive probability approaches one for any given pair of error exponents.

The Bayesian average error of quantum hypothesis testing can also be analyzed when there are multiple hypotheses, a setting often known as quantum state discrimination, and its optimal error exponent has been established in Ref. [15]. In my works with Mishra, Mosonyi, and Wilde [16, 17], we investigated a variation of the discrimination task called *quantum state exclusion*, where the goal is no longer to guess the true identity of an unknown state but to rule out a false identity from multiple given hypotheses. Apart from its information-theoretic relevance, quantum state exclusion has also found use in the ontological interpretation of quantum states [18]. In our work, we derived the tightest known single-letter and efficiently computable converse bound on the error exponent of quantum state exclusion, and furthermore, of quantum channel exclusion. As special cases, our result leads to an exact characterization of the error exponent of classical channel exclusion, generalizing prior results on binary classical channel discrimination [19], and provides a simpler derivation of the error exponents of classical state exclusion [20] and of binary classical state discrimination [12].

1.3 Thermodynamics and quantum side information

Quantum side information is fundamentally different from classical side information, as exemplified by the fact that conditional entropy can be negative in the presence of quantum side information [21], an impossibility when the side information is classical. Negative conditional entropy challenged the traditional understanding of information [22], and in resolving this, various operational meanings of it have been discovered, in contexts ranging from communication [22, 23] and the uncertainty principle [24] to thermodynamics [25]. The study of quantum side information has been an epitome of how the analytical approach of quantum information theory contributes fundamental conceptual insights to foundational questions in quantum theory such as the characterization of nonclassicality.

In my work with Gour and Wilde [26], we developed a resource theory that sharpens the quantitative relationship between thermodynamics and quantum side information, formulated in the operational language of feedback control. The study of feedback control [27, 28] is essential to the modern understanding of Maxwell's demon [29], a paradox that can essentially be attributed to unaccounted-for side information. However, previous studies of feedback control [28], even for quantum systems [30, 27, 31], do not effectively incorporate quantum side information, as the side information being fed back is assumed to be collected by measurements. This fails to capture a fully quantum Maxwell's demon, who is capable of acquiring, processing, and feeding back quantum side information *coherently* [32]. With our resource theory, we filled this gap and derived fundamental no-go limits for thermodynamic control in the presence of coherent quantum feedback. We established axiomatically tight bounds on the one-shot work of formation and extractable work given quantum feedback, in terms of smoothed generalized mutual informations. In the special case of trivial Hamiltonians, this provides precise thermodynamic meanings for the smoothed conditional min- and max-entropies and their negativity. In the asymptotic limit, these bounds give rise to a generalized second law of thermodynamics with quantum feedback, featuring a conditional version of the Helmholtz free energy, and we furthermore showed that the generalized law remains consistent with the traditional second law in any complete cyclic process. Apart from its thermodynamic implications, our work also contributed several technical insights to information theory: it proposes a new way of smoothing information measures, slightly different from the commonly adopted convention [33, 34], and it also led to an affirmative proof of an open conjecture in the program of axiomatizing quantum conditional entropies raised in Ref. [35]. It also lays the foundation for the axiomatization of information measures that involve quantum side information, nontrivial Hamiltonians, and error tolerance, thus in part speaking to open questions raised in Ref. [36, 37].

1.4 Other works

My work with Chitambar [38] studied the role of quantum memories in programmable quantum instruments. We argued that the demand for quantum memories in such a programmable device is equivalent to the incompatibility of the set of quantum instruments it implements, and we proposed a semi-device-independent test for it. Our operational framework unifies those for measurement incompatibility [39] and EPR steering [40].

My other work with Chitambar [41] proposed a generalized conditional min-entropy as an uncertainty measure for operationally restricted agents. With the proposed definition, we provided information-theoretic interpretations for one-shot resource convertibility and for resource robustness measures in general convex quantum resource theories for states and channels, confirming a conjecture in Ref. [42] on the connection between resource measures, information measures, and operational tasks.

2 Future directions

2.1 Extension of past works

Here are some examples of the directions that naturally extend, or are directly related to, my past works.

- *Operational and/or axiomatic interpretations of information measures.*

  One of the main quests of quantum information theory is to determine the ultimate limits of quantum information-processing tasks, and it is a rare delight whenever such a limit can be pinned down precisely in terms of analytic information measures. Here the benefit flows in the other direction as well: these limits in turn endow the relevant information measures with precise operational meanings, providing moral guidance for the technical investigation of such measures. A great example of this is how the quantum Stein's lemma [43, 44] singles out the Umegaki relative entropy among all plausible quantum generalizations of the Kullback–Leibler divergence by resorting to quantum hypothesis testing. Parallel to the operational approach, one can also gain principled guidance for studying information measures by following an axiomatic approach, with the aim of reconstructing known information measures by demanding certain essential properties.

  The triad between the operational, axiomatic, and analytic approaches to information measures has been at the forefront of my past works, and I have been incredibly fortunate to contribute in this regard to the understanding of various information measures, including the smoothed conditional min- and max-entropies [26], the projective [4] and Doeblin channel informations [1], etc. I expect to continue this line of work and hope to discover more such beautiful coincidences. The operational and/or axiomatic interpretations of many prominent information measures remain unresolved, such as the geometric family [45, 46] (including the Belavkin–Staszewski [47]) and the log-Euclidean family [48] of relative entropies, and a quantum generalization of the lautum information [49]. In addition, the operational and/or axiomatic approaches could lead to the discovery of new information measures; for instance, it is expected that conceptually novel multivariate distinguishability measures will be needed for exactly determining the optimal error exponent of quantum state exclusion [20, 16].

- *Information processing with a small failure probability.*

  In my past work with Regula [13], we demonstrated that the exponent tradeoff between the different types of errors in binary quantum hypothesis testing can be significantly improved if one tolerates a vanishingly small probability of being inconclusive. Owing to the intimate connection between (composite) quantum hypothesis testing and resource distillation [50, 51, 52], it is natural to expect that a similar improvement can be established for the latter as well. Specifically, the tradeoff in resource distillation is between the yield and the fidelity; for a given yield, we expect that the fidelity would approach one at a faster pace under protocols with a vanishingly small failure probability than under protocols that strictly always succeed. Moreover, unlike in quantum hypothesis testing, we have strong reasons to believe that the improvement in the fidelity of resource distillation persists even if the yield approaches zero. Overall, this would provide practical means of overcoming traditional limits on the reliability of quantum information processing and open up new possibilities for demonstrating similar improvements in other applications such as parameter estimation and communication.

- *Thermodynamics, economics, and natural selection.*

  There are many shared structures between the mathematics underlying thermodynamics and that underlying economics. From an information-theoretic perspective, one such resemblance is between Kelly gambling [53] and work extraction. Specifically, both the average growth rate of wealth in gambling and the average extracted work via quenching (i.e., by changing the Hamiltonian without changing the actual state, which is different from applying a free operation to the actual state as in resource-theoretic approaches) are given by a relative-entropy difference. Moreover, it has recently been identified [54] that there is an isomorphism between wealth growth in gambling and work extraction in thermodynamics even at the microscopic level, with the inverse odds corresponding to the initial Gibbs weights and the allocation of bets corresponding to the final Gibbs weights.

  I have a preliminary result showing that that both tasks are precisely related to hypothesis testing in the one-shot regime; specifically, given a success probability, the optimal deterministic growth rate of wealth and the optimal deterministic extractable work via quenching are both equal to a variant of the hypothesis-testing relative entropy that optimizes only over projective tests. If the gambler can "parlay" their bets with another bet on a random seed, then the usual hypothesis-testing relative entropy is recovered. It is also known that side information provides an average advantage quantified by the mutual information in Kelly gambling [55], as is the case in work extraction [28]. As a one-shot generalization, my preliminary result shows that the relevant quantities are precisely linked to the smoothed generalized min-mutual information, a quantity that arises in my past work [26] on the resource theory of thermodynamics with quantum side information. Another fact worth mentioning is that the balance between growth and risk in gambling is essentially equivalent to the tradeoff between the two types of error exponents in binary hypothesis testing, where the Rényi relative entropies make an appearance [56].

  Lastly, in a community of economic enterprises or biological organisms, evolution via natural selection is highly contingent on the wealth growth or the work extraction of the individuals [57]. One could likewise expect that insights in this regard can be gained from concepts and techniques developed in information theory.

2.2 Possible explorations

One of the credos I try to uphold is that every one of my research projects should fall into at least one of the following two categories: either it is a better work than all of my previous works, or it introduces me to something new, exposing me to new ideas, concepts, and/or techniques. I am fortunate to be able to say that I have indeed held on to this credo so far. Looking into the future, it dictates that the scope of my research will never stop expanding, and I fully embrace such opportunities when they arise. Here are some examples of the directions that I can imagine myself working on in the future.

- *Quantum learning.*

  A primary goal of my research is to understand how nature works. To better achieve this goal, it is certainly instructive to understand "understanding" itself, and this brings us to the realm of learning theory. Similar to information theory, learning theory also concerns ultimate limits, in particular for learning tasks, and there still seems to be much fundamental work to do towards a quantum theory of learning, where the machine and/or the data can be quantum, especially under various practical resource constraints. Another direction that stands out to me is reinforcement learning with quantum agents [58] and/or environments [59]. Due to the nature of the questions it is faced with, reinforcement learning in a quantum setup likely has deep conceptual intersections with quantum stochastic processes [60], quantum causal models [61], non-Markovianity [62], and feedback control [63]. It is also part of the goal of my research to distill high-level methodological wisdom from the concrete questions I work on and to guide my everyday life. As a big part of a good life is learning, studying how to learn effectively from an analytical perspective is certainly a good fulfillment of this goal.

- *Quantum causality.*

  Causality is what separates understanding from knowing, and wisdom from intelligence. One can argue that it is the whole point of science. However, causality is not fully understood in the quantum world. For instance, unlike in classical processes, the relation between causal constraints and compositional structures in quantum processes is rather ambiguous due to noncommutativity, and it can become even more complicated due to the possibility of indefinite causal structure [64, 65] or cyclic causal influence [66]. Quantum causality has also been believed to be relevant to understanding quantum gravity.

  There is a fruitful history of quantum causality and quantum information theory inspiring each other. In one direction, early work on the causal structure of quantum processes [67, 68, 69] and networks [70] has laid the foundation for the study of information-processing tasks under causal constraints [71, 72] or using complex strategies [73, 19, 74]; in the other direction, the analytical approach of quantum information theory has supplied profound insights to the study of quantum causality [75, 61] as well. Likewise, in the majority of my own works [38, 4, 16, 26, 1], meticulous reasoning about the causal and/or compositional structure of the relevant operations is required before the information-theoretic analysis can proceed, and this has produced structural results that may be of independent interest even from a purely causality perspective. Indeed, in the study of quantum processes, unification between axiomatic and constructive approaches (namely, identifying equivalences between simple properties and structure decompositions [67, 69, 70, 76, 77]) has been a recurring theme, one that has appealed to me tremendously since my first acquaintance with quantum information theory. For all these reasons, an even deeper dive into quantum causality and its interplay with quantum information theory would be very welcome to me.

- *The measurement problem and nonclassical correlations.*

  The extended-Wigner's-friend paradox [78] is one of the biggest recent breakthroughs in quantum foundations. It formalizes an outright logical inconsistency within quantum theory and renders the troubling implication of the measurement problem undodgeable in a precise sense. It has also inspired the discovery of a class of neither-classical-nor-quantum correlations known as local friendliness [79], which employs a probabilistic (as opposed to possibilistic in Ref. [78]) notion of such paradoxes.

The original paradox [78] can be understood as arising from a thought experiment in which the Hardy state [80] is measured. In an attempt to understand what is distinctive about the Hardy state that enables the paradox, I proved the following result (although I later became aware that the result was subsumed by an earlier work [81]): for a quantum state shared among an arbitrary number of parties, an extended-Wigner's-friend-like thought experiment can be constructed, which involves measuring the state and from which probabilistic logical inconsistency can arise, if and only if the state is multipartite-nonlocal in the Bell sense. A future question would be to determine the exact class of states for which possibilistic inconsistency can arise. In addition, my proof for the probabilistic case motivates the definition of a hierarchy of linearly constrained multipartite correlations, interpolating between correlations admitting local hidden variables and correlations satisfying local friendliness. Likewise, the answer to the question in the possibilistic case would shed light on the understanding of possibilistic local friendliness [82].

In my past work with Lloyd and Wilde [1], we established the ultimate limit of retrocausal communication. I explored applying our optimal communication protocol to noisy CTCs arising in the vicinity of black holes under final-state-projection models, yet became alerted by the fact that such a protocol would involve performing measurements inside black holes and thus could be trickier than it appears, after coming across Ref. [83]. The implications of performing measurements inside black holes and their consequence on exploiting black holes for retrocausal communication remain to be understood.

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